

LUNA ZEA-REDONDO, Ph.D.

Computational Biologist | AI & Machine Learning for Life Sciences

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SUMMARY

Computational Biologist with **10+ years** of experience applying quantitative and statistical approaches to complex biological systems, spanning cancer genomics to neuroepigenetics. **PhD in Systems Biology** with strong expertise in single-cell, multi-omics, and time-course data analysis. Now extending this foundation with machine learning and **production ML engineering**, bringing computational rigor to challenges at the intersection of **AI and Life Sciences**.

PROFESSIONAL EXPERIENCE

Senior Research Scientist | Computational Biology

Berlin Institute for Medical Systems Biology (MDC-BIMSB)

Oct 2025 – present

Berlin, Germany

- Brought a large-scale single-cell multi-omics study, involving **multi-timepoint biomarker** data, to **first-author manuscript submission**, leading the analysis decisions and scientific narrative in coordination with senior stakeholders.
- Initiated and managed two **collaborative follow-up projects** stemming from my PhD, including experimental **validation** of a key epigenetic finding, across **four international cross-functional teams**.
- **Mentor** master's and PhD students and serve as **peer reviewer** for >5 international scientific journals, critically evaluating experimental design, analysis and strength of scientific conclusions.

Ph.D. Researcher | Computational Biology

Berlin Institute for Medical Systems Biology (MDC-BIMSB)

Jan 2019 – Oct 2025

Berlin, Germany

- **Designed and led** a single-cell multi-omics time-course end-to-end: experimental design, hands-on library generation at the bench, and computational analysis (18 samples, >150,000 cells, 6 time points), applying **time-course statistical modeling** to uncover the transcriptomic and regulatory basis of cellular memory following a single drug exposure in specific *in vivo* neuronal populations.
- **Co-led** a collaborative multi-omics project culminating in a **co-first-author *Nature* (2021) publication** (>200 citations), integrating scRNA-seq, scATAC-seq, and 3D genome topology across brain cell types.
- **Generated and analyzed >10 large-scale genomic** (single-cell RNA and ATAC, bulk ATAC and ChIP-seq) datasets, contributing to five further publications through collaborations with internal and external teams
- **Awarded** a competitive DFG-funded binational PhD scholarship (Humboldt–Duke); selected for talks at international conferences, receiving Outstanding Research Award and travel fellowship prizes.

Junior Scientist | Cancer Genomics & Computational Biology

New York University Medical School (NYUMS)

Feb 2017 – Aug 2018

New York, US

- Characterized the regulatory impact of a candidate disease biomarker in a blood cancer, **generating and analyzing >20 ChIP-, RNA-, and ATAC-seq datasets** from primary patient blood samples and human cell lines.

- **Analyzed > 10 ChIP-seq and ATAC-seq datasets** for studies on inflammation and leukemic transformation contributing to **high-impact publications** (*Nature Immunology* and *Cell Stem Cell*).
- Contributed analysis and figures supporting a **\$2.5M NIH R01** grant application on enhancer regulation in leukemia.

Research Fellow | Molecular Biology & Functional Genomics

European Molecular Biology Laboratory (EMBL)

Max-Planck-Institute for Molecular Genetics (MPIMG)

Oct 2015 – Sep 2016

Rome, Italy

Berlin, Germany

- Conducted **hands-on molecular biology** work bridging wet-lab and computational expertise across two leading European research institutes, gaining foundational experience in experimental design and execution.

SKILLS

- **Machine Learning & AI:** Scikit-learn, supervised/unsupervised learning, statistical modeling; deep learning fundamentals; LLM & agentic-AI workflows (Agentic RAG, Tool Calling, LLM-as-a-judge, embeddings).
- **Computational Biology & Multi-Omics:** RNA-seq, ATAC-seq, ChIP-seq; epigenomic and transcriptomic analysis; multi-omics integration; QC and validation; time-course modeling; clinical domain knowledge.
- **Data Engineering & Cloud:** Linux/HPC; Docker, Elasticsearch (Vector DB), Terraform, GCP, BigQuery, dbt, DuckDB; Apache Spark, Kafka, Airflow, data warehousing; ETL/ELT pipeline design.
- **Programming & Frameworks:** Python, R, SQL, Shell, Git, Streamlit.
- **Communication & Leadership:** Cross-functional collaboration, project planning, mentoring, scientific writing, peer review for international journals, conference presentations.

EDUCATION

Data Science & AI Intensive Program (*in progress*) • Constructor Academy • online • **2026**

PhD in Systems Biology • Humboldt University of Berlin • Berlin, Germany (*Summa cum laude*) • **2025**

MSc in Bioinformatics • Autonomous University of Barcelona • Barcelona, Spain • **2017**

BSc in Biochemistry and Molecular Biology • University of Córdoba • Córdoba, Spain • **2015**

SELECTED PROJECTS

Full details, code, and interactive demos at: red-moonx.github.io

- **Athletica** Agentic RAG system utilizing Dense Vector Search and LLM tool-calling to provide accurate, hallucination-free sports nutrition guidance based on PubMed literature. *Python, OpenAI, Elasticsearch, Streamlit*.
- **Clinharmonia:** Clinical data pipeline harmonizing multi-source oncology cohort data (TCGA-BRCA) into ML-ready analytical marts. *dbt, DuckDB, SQL*
- **Cetacean Global Biomonitor:** Data engineering pipeline transforming 11M+ biodiversity records into a cloud data warehouse for geospatial analysis. *Terraform, Airflow, BigQuery*

SELECTED PUBLICATIONS

1. Zea-Redondo, L.* et al. (2021). Cell-type specialization is encoded by specific chromatin topologies. *Nature* (*shared co-first*).
2. Muto, T. et al. (2022). TRAF6 functions as a tumor suppressor in myeloid malignancies. *Cell Stem Cell*
3. Guillaumot, M. et al. (2019). The E3 ubiquitin ligase SPOP controls resolution of systemic inflammation. *Nature Immunology*.